

WELLSENSE · DESIGN DECISIONS REGISTER

WellSense design decisions register

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DOCUMENT CONTROL

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Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/wellsense		
Source	docs/06-design-decisions.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.1	2026-10-02	Amish Chadha	Register opened with the build plan; budget treated as a value-engineering target

WellSense design decisions register

Every design decision still to be made, and every decision made, in one place. Each decision is argued in full in its decision record under `docs/decisions/` ; this register is the index Amish works from. The build plan (`docs/05-build-plan.md`) describes the design as it stands and does not list open decisions.

■ Open decisions

#	Decision needed	Options	Recommendation	Affects in the build	Source
1	Review the design-for-construction changes P1 to P11 (split seal plate, collar, drilled tube, tube cap with cross bolt and grip, flexible tail, bent conduit on saddles, junction box plate with V-blocks, box entries and internal plate, barometric housing, FieldNode lead, post cap)	Accept as made; or ask for changes	Accept	The whole build plan	WLS-DDR-004, made under Amish's 2026-09-30 instruction; open for his review
2	Locking of the wellhead and junction box (R16)	(a) security-head screw on the rim band, padlock through the cross bolt's head, padlockable hasp kit on the box; (b) lockable steel cover over the wellhead	(a), decided after the first site visit	Seal plate, tube cap, junction box	WLS-DDR-004, A1; WLS-PRC-001 open questions
3	First partner and region for co-design	An Indian farmer group, an African handpump programme or a US groundwater agency	None yet	Design well (casing, riser and pump cable sizes); LoRaWAN band of the FieldNode core	WLS-DDR-001, O1
4	Battery-only FieldNode variant without the panel	Keep the standard core; or a battery-only core	None (for the FieldNode project)	FieldNode core	WLS-DDR-001, O2
5	FieldNode sensor port pin assignment	Pinout of port A agreed with the FieldNode project; WellSense assumes 12 V rail, ground and I2C	None yet (FieldNode decision)	FieldNode lead wiring (build plan section 3.5.1)	FND-DDR-001, O2
6	FieldNode core above 45 °C ambient (R15 asks for 55 °C)	FieldNode confirms the shielded core at 55 °C; or R15 restated to FieldNode's range	None yet (FieldNode decision)	FieldNode sun shield at hot sites	REVIEW.md, cross-repo actions
7	How the FieldNode geometry in cad/vendor is kept up to date	(a) re-export it whenever FieldNode's general arrangement is revised; (b) a shared library repo	(a)	Model, drawings and pictures of the FieldNode core	WLS-DDR-004, A2
8	Photoreal render layout: post drawn 330 mm from the well and 350 mm lower than installed	Accept for renders only; or render the installed layout	Accept for renders only	Renders only; not the build	REVIEW.md, 2026-09-26, render item 1

#	Decision needed	Options	Recommendation	Affects in the build	Source
9	Render grouping of the post, tube and cable for the exploded render	Accept; or group as the BOM	Accept	Renders only	REVIEW.md, 2026-09-26, render item 5

Render items 3 (conduit route at the tube cap) and 4 (seal plate gasket) of the 2026-09-26 review are overtaken by WLS-DDR-004: the conduit now ends in a flexible tail and the gasket is a separate ring under the plate. The renders need redoing on Amish's Mac in any case.

■ To confirm when parts are bought

#	What to confirm	Why it matters	Source
1	Drinking-water certificates for the probe, the vented cable, the HDPE sheet and the EPDM (gasket and wraps)	R9 is not met without them	WLS-REQ-001, R9; WLS-DDR-004
2	The riser and pump cable diameters at the well	They set the seal plate's split holes (designed for a 42.2 mm riser and a 12 mm cable)	WLS-DDR-004, P6
3	The junction box model, its boss spacing (80 x 110 mm assumed) and its lug kit	They set the internal plate holes and the lug holes in the plate	WLS-DDR-004, P3
4	The shaft collar's rated axial grip on PVC pipe, at least 264 N	It carries the tube at 60 m	WLS-CAL-001, [D5]
5	The cable support grip suits 6 to 8 mm cable and carries at least 35 N	It carries the probe and cable at 60 m	WLS-CAL-001, [D6]
6	The probe body is 22 mm or less, and its supplier's price	2.3 mm radial clearance in the tube; the BOM assumes the 26 to 28 mm class price	WLS-DDR-002; BOM line 1
7	The flexible tail's connector fits a 22.5 mm hole in a 1 in PVC cap, with its locknut inside	The cap is the connector's only support	WLS-DDR-004, P5 and P8

■ Value engineering

Value-engineering target: USD 200 (a hypothetical control target, not a limit). Estimated cost of the constructable design: USD 244.40 for the WellSense parts at the 30 m design depth (USD 44.40 over the target); USD 383.40 with the FieldNode core, which is costed in the FieldNode repo. Main cost drivers and savings worth trying:

- The largest lines are the transducer (USD 50), the access tube (USD 31 at 30 m), the post with its plate, V-blocks and bands (USD 32), the vented cable (USD 26.40 at 30 m), the seal plate set (USD 23) and the conduit set (USD 20). Cable and tube scale with depth at USD 1.80 a metre.
- Making the design constructable added USD 46.80: the seal plate set (+8), tube cap and grip (+4), junction box lugs, glands and internal plate (+6), terminal strip (+1), junction box plate and V-blocks (+10), flexible tail, saddles and hub (+10), the FieldNode lead (+7) and 1 m more cable (+0.80).

- Savings worth trying: fit the conduit set only where livestock or tampering is likely (USD 20); leave the access tube out where no pump shares the casing (USD 31 at 30 m, already the rule); mount the barometric module inside the junction box, which breathes through its desiccant, instead of in its own housing (about USD 4); buy the junction box with a pole-mount kit if one is offered at less than the plate, V-blocks and bands (up to USD 10); cut the seal plate from offcuts shared across several wells.

■ Decisions made

Date	Decision	Decided by	Record
2026-09-25	TRL 2 items D1 to D8: FieldNode core costed in its own repo; vented 4 to 20 mA transducer; 0 to 10 m range; own access tube where a pump shares the casing; 15 min readings, 1 min in pumping tests; quarterly manual tape check; levels only, community controls sharing; FieldNode as the core	Amish: "i accept all your recommendations, go with them across all repos."	WLS-DDR-001, WLS-DDR-002
2026-09-25	TRL 3 items A9 to A13: design depth 30 m; 1 in tube with a probe of 22 mm or less; galvanized conduit over the surface cable; FieldNode sun shield at hot sites; pumping tests send every 30 min on a public network	Amish, same instruction	WLS-DDR-002
2026-09-26	<code>budget_usd</code> set to \$200	Amish: "i approve all the budget items."	WLS-DDR-002, N1
2026-09-27	FieldNode panel tilt corrected to face the way the node faces	Amish: "resolve the challenges for ConePro, BridgePulse, Grainguard and WellSense."	WLS-DDR-003
2026-09-30	Make the design physically buildable while drawing the build plan	Amish: "If you are realising that the design cannot be built as per concept - fix the design assumptions to match and be physically feasible as you draw the illustrations." Changes made under this instruction; open for his review (open decision 1)	WLS-DDR-004
2026-10-01	<code>budget_usd</code> is a value-engineering target, not a limit	Amish: "the budgets are a hypothetical control target to ensure we are thinking along a value engineering lens. its ok to ensure wording reflects that the hypothesis budget was x - the real cost being accrued is y"	STANDARDS section 18; this register